Homework 9

Due: Monday, November 16 at 11:59 p.m. PT

MATH 112, Fall 2026

Instructions

- As part of your preparation for Exam 2, you must also submit the Exam 2 Concept Map as a separate assignment on Gradescope by **Wednesday, November 11 at 11:59 p.m. PT**, before the exam on Thursday, November 12. The concept map is graded for completion.
- Write your full name, "Homework 9," and the date at the top of the first page.
- Show all work and explain your reasoning. Write in complete sentences.
- Typeset your solutions in LaTeX if you can.
- Submit a single PDF file to Gradescope under the assignment "Homework 9." Submit the concept map separately under its corresponding assignment.
- Use Gradescope to electronically match problems to pages in your PDF.
- Questions? Email the instructor or come to office hours; see Moodle.
- You are strongly encouraged to work together. Write up your own solutions.

Be sure to show your work. Organize it so that another MATH 112 student who has not seen the problems can follow your reasoning from line to line. Solutions that contain only an answer, without justification or supporting work, will not receive full credit.

Assignment 1 (graded for completion; due on Wednesday 11/11 at 11:59 PT)

As part of reviewing for Exam 2, draw a concept map illustrating connections between the main concepts, definitions, and theorems we have studied so far. **Emphasize the material from Week 5 onward**, while including relevant concepts from earlier weeks to show how the ideas connect. You may use the example concept map on Moodle as a guide, but your map does not need to follow its structure. Try to label arrows to explain how concepts and results are related.

Upload your concept map to its corresponding **separate assignment on Gradescope by Wednesday, November 11 at 11:59 p.m. PT**. Recall that Exam 2 is on Thursday, November 12. The homework problems below are due November 16 at 11:59 PT (note the unusual day- Monday instead of a Friday).

Problem 1 (15 points)

Geometric sequences, boundedness, and the squeeze theorem.

(a) For each of the following geometric sequences, determine whether it converges. If it converges, find its limit; otherwise, describe its limiting behavior (including the behavior of its modulus, where relevant). Justify your answers using the theorem on geometric sequences.

$$x_n = \left(\frac{1+i}{3}\right)^n, \qquad y_n = (-i)^n, \qquad z_n = (1+i)^n.$$

(b) Let (a_n) be a convergent sequence of complex numbers. Show that

$$\frac{a_n}{2^n} \longrightarrow 0.$$

(c) More generally, let $r \in \mathbb{C}$ with $|r| < 1$. Show that

$$a_n r^n \longrightarrow 0.$$

(d) Use an appropriate result to compute

$$\lim_{n\to\infty} \frac{(-1)^n + 3}{2^n}.$$

Problem 2 (20 points)

Computing limits: ratio criterion, root criterion, and comparison.

For each of the following sequences, determine its limiting behavior. If it converges, find its limit; if it diverges to $+\infty$, justify this using an appropriate criterion or comparison theorem.

(a) $a_n = \dfrac{n^4}{3^n}$.

(b) $b_n = \dfrac{n!}{4^n}$.

(c) $c_n = \left(\dfrac{2n+1}{3n-1}\right)^n$.

(d) $d_n = n + \dfrac{n}{2}(-1)^n$.

Problem 3 (15 points)

Monotone convergence and the ratio criterion.

Fix $\alpha > 0$ and define

$$a_n = \frac{1+n\alpha}{(1+\alpha)^n}, \qquad n \in \mathbb{N}.$$

(a) Show that (a_n) is decreasing.

(b) Use the Monotone Convergence Theorem to prove that (a_n) converges.

(c) Determine its limit using the ratio criterion.

Problem 4 (20 points)

Radicals and the Monotone Convergence Theorem

Consider the sequence defined recursively by

$$a_1 = \sqrt{3}, \qquad a_{n+1} = \sqrt{3 + a_n}, \quad n \in \mathbb{N}.$$

(a) Show that $0 < a_n < 3$ for every $n \in \mathbb{N}$.

(b) Prove that (a_n) is increasing.

(c) Use the Monotone Convergence Theorem to show that (a_n) converges.

(d) Determine its limit.

Problem 5 (15 points)

A recursive sequence converging to $\sqrt{2}$.

Consider the recursively defined sequence

$$a_1 = 2, \qquad a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right), \qquad n \in \mathbb{N}.$$

(a) Prove that $a_n \geq \sqrt{2}$ for every $n \in \mathbb{N}$. *Hint:* For $x > 0$, consider

$$\frac{1}{2}\left(x + \frac{2}{x}\right) - \sqrt{2}.$$

(b) Show that (a_n) is decreasing.

(c) Use the Monotone Convergence Theorem to show that (a_n) converges.

(d) Determine its limit.

Problem 6 (15 points)

Cauchy sequences and the distance between terms.

(a) Prove that every Cauchy sequence of complex numbers is bounded, using only the definition of a Cauchy sequence.
Hint: Recall how we proved that every convergent sequence is bounded.

(b) Suppose that (a_n) is a convergent sequence of complex numbers. Prove that, for every *fixed* $m \in \mathbb{N}$,

$$\lim_{n\to\infty} (a_{n+m} - a_n) = 0.$$

Is the converse true? That is, if a sequence satisfies this condition for every fixed $m \in \mathbb{N}$, must it converge? Must it be a Cauchy sequence? Prove your answer or give a counterexample.
Takeaway: Compare carefully what the condition above says with the definition of a Cauchy sequence. Explain the difference between requiring two terms to be close when their indices are a *fixed distance apart* and requiring *any two sufficiently far-out terms* to be close. Where does the order of the quantifiers matter?